Broken Web App Challenge

Step-by-step resolution guide (AWS CloudFormation / EC2 / ALB / Auto Scaling)

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1. What the lab is about

A CloudFormation template builds a VPC with two public and two private subnets, an Auto Scaling Group (ASG) of Apache web servers, and an Application Load Balancer (ALB). Visiting the ALB DNS name fails. The goal is to fix the **live environment** (not the template) so the web page loads. There are exactly **four** issues.

Traffic path to keep in mind:

```
Internet -> ALB (public subnets, ALB SG) -> Listener :80 -> Target Group (health check)
         -> EC2 instances in ASG (Web SG, must allow port 80 from ALB SG)
```

2. The four issues at a glance

#	Component	What is broken	Fix
1	ALB subnets	ALB sits in the two **private** subnets (no route to the Internet Gateway), so it is unreachable from the internet.	Move the ALB to the two **public** subnets.
2	Target group health check	Health check path is `/healthcheck.hmtl` (typo). The file is `healthcheck.html`, so targets fail health checks.	Change path to `/healthcheck.html`.
3	Web security group (websg)	Inbound rule allows only port **22** (SSH) from the ALB SG. Port **80** is not allowed, so the ALB cannot reach Apache.	Add inbound HTTP 80 from the ALB security group.
4	ASG to Target Group	The Auto Scaling Group is not attached to the target group, so no instances are registered behind the ALB.	Attach the target group to the ASG.

3. Step 0 - Deploy the lab

1. Sign in to the AWS Management Console and set the Region (top-right) to **US East (N. Virginia) us-east-1**. This is mandatory.
2. Open **CloudFormation** > **Create stack** > **With new resources (standard)**.
3. Choose **Amazon S3 URL** and paste:

```
https://cloud-mastery-bootcamp.s3.amazonaws.com/lab-guides/
solutions-architecture/broken-web-app-challenge.json
```

4. Click **Next**. For **Stack name** enter your first name (e.g. *sukhada*). Keep default parameters (t3.micro, 2 instances).
5. Click **Next** twice. On the review page, tick **I acknowledge that AWS CloudFormation might create IAM resources** and click **Submit**.
6. Wait until the stack status is **CREATE_COMPLETE** (about 3-5 minutes). Open the **Outputs** tab and copy the ALB URL.
7. Open the URL in a browser. It will fail (timeout / 503 / 504). This confirms the environment is broken.

> **Tip:** Only fix things in the live resources (EC2, ELB, ASG consoles). Do not edit and update the CloudFormation stack.

4. Fix 1 - Move the ALB into the public subnets

Why: The ALB is internet-facing but was created in private subnets. Private subnets route to the private route table, which has no 0.0.0.0/0 route to the Internet Gateway, so the load balancer cannot be reached from the internet.

Console steps

1. Go to **EC2** > **Load Balancers** and select the ALB created by your stack.
2. Open the **Network mapping** tab and click **Edit subnets**.
3. For **us-east-1a** select the public subnet **10.0.0.0/24** (name: public-10.0.0.0-us-east-1a) instead of the private one.
4. For **us-east-1b** select the public subnet **10.0.2.0/24** (name: public-10.0.2.0-us-east-1b) instead of the private one.
5. Click **Save changes**.

> **Careful:** The tag on the second private subnet (10.0.3.0/24) is wrongly named 'private-10.0.1.0-us-east-1a'. Identify subnets by their **CIDR**, not just their Name tag. Public subnets are 10.0.0.0/24 and 10.0.2.0/24. To double check, open VPC > Route tables > public-rt > Routes; it must contain 0.0.0.0/0 to the igw.

If the console will not let you change the subnets, delete the ALB and re-create it (internet-facing, public subnets, ALB security group, listener HTTP:80 forwarding to the existing target group).

CLI alternative

```
aws elbv2 set-subnets --region us-east-1 \
  --load-balancer-arn <ALB_ARN> \
  --subnets <PUBLIC_SUBNET_1A_ID> <PUBLIC_SUBNET_1B_ID>
```

5. Fix 2 - Correct the target group health check path

Why: User data creates `/var/www/html/healthcheck.html`, but the target group checks `/healthcheck.hmtl` (letters swapped). Apache returns 404, targets are marked unhealthy and the ALB returns 503.

Console steps

1. Go to **EC2** > **Target Groups** and select the target group of your stack.
2. Open the **Health checks** tab > **Edit** (or Actions > Edit health check settings).
3. Change **Health check path** from `/healthcheck.hmtl` to `/healthcheck.html`.
4. Click **Save changes**.

CLI alternative

```
aws elbv2 modify-target-group --region us-east-1 \
  --target-group-arn <TARGET_GROUP_ARN> \
  --health-check-path /healthcheck.html
```

6. Fix 3 - Allow HTTP (port 80) from the ALB to the web servers

Why: The web server security group (**websg**) only allows TCP 22 from the ALB security group. The ALB needs to reach the instances on TCP 80, otherwise health checks and requests time out.

Console steps

1. Go to **EC2** > **Security Groups** and select the group named **websg**.
2. Open the **Inbound rules** tab > **Edit inbound rules** > **Add rule**.
3. Type: **HTTP** (TCP, port 80). Source: **Custom** and choose the ALB security group (**albsg**, it starts with sg-).
4. Optional clean-up: delete the SSH (22) rule from the ALB SG since an ALB never SSHs to instances.
5. Click **Save rules**.

CLI alternative

```
aws ec2 authorize-security-group-ingress --region us-east-1 \
  --group-id <WEBSG_ID> --protocol tcp --port 80 \
  --source-group <ALBSG_ID>
```

Referencing the ALB security group as the source (instead of 0.0.0.0/0) is best practice: only the load balancer can reach the instances on port 80.

7. Fix 4 - Attach the Auto Scaling Group to the target group

Why: The ASG launches the instances, but nothing registers them in the ALB target group. The target group shows *no registered targets*, so the ALB has nowhere to forward traffic.

Console steps

1. Go to **EC2** > **Auto Scaling Groups** and select the ASG (instances are tagged 'autoscalinginstances').
2. Open the **Integrations** tab (called 'Load balancing' in older consoles) > **Load balancing** > **Edit**.
3. Tick **Application, Network or Gateway Load Balancer target groups**.
4. Select the stack's target group from the drop-down and click **Update**.

CLI alternative

```
aws autoscaling attach-load-balancer-target-groups --region us-east-1 \
  --auto-scaling-group-name <ASG_NAME> \
  --target-group-arns <TARGET_GROUP_ARN>
```

Alternative: you can instead register the two running instances manually in Target Groups > Targets > Register targets. Attaching the ASG is the proper fix because new instances launched by scaling are registered automatically.

8. Verify the fix

1. Go to **EC2** > **Target Groups** > your target group > **Targets** tab. Wait 1-2 minutes until both instances show **healthy** (they need 2 consecutive successful checks; interval is 30 s).
2. Open the ALB DNS name from the CloudFormation **Outputs** tab (use **http://**, not https).
3. The web page should load and show the Availability Zone of the instance. Refresh a few times; the AZ may alternate between us-east-1a and us-east-1b, which proves load balancing works.

9. Quick troubleshooting table

Symptom	Likely cause	Check
Browser times out	ALB not in public subnets / ALB SG blocks	Fix 1; ALB SG allows inbound 80 from 0.0.0.0/0
503 Service Unavailable	No healthy / no registered targets	Fix 4, then Fix 2
504 Gateway Timeout	Instances unreachable on port 80	Fix 3 (websg inbound 80 from albsg)
Targets 'unhealthy' (404)	Wrong health check path	Fix 2 (/healthcheck.html)

10. Clean up (avoid charges)

1. Open **CloudFormation**, select your stack and click **Delete**.
2. Wait for **DELETE_COMPLETE**. If it fails, delete the ALB / manually added rules first, then retry.

11. What you learned

- Public vs private subnets are defined by their route to an Internet Gateway; an internet-facing ALB must be in public subnets.
- ALB health checks must point to a path that returns HTTP 200 (a typo makes every target unhealthy).
- Security groups chain: Internet > albsg (80) > websg (80 from albsg).
- An ASG must be attached to a target group for its instances to receive ALB traffic.

- Troubleshoot in order: network path, listener/target group, health checks, security groups, registration.